

Cloud Computing & DevOps– Task 1

Host a Static Portfolio on AWS S3 + Serve via CloudFront (HTTPS)

Objective:

Learn the basics of cloud object storage, static hosting, and a simple CDN configuration. You will create a polished single–page portfolio (HTML/CSS) and host it on AWS S3, secure it with CloudFront (HTTPS), and optionally map a custom domain.

Why this task?

This combines practical cloud knowledge (S3, IAM, bucket policies) with DevOps–adjacent skills (CDN, caching, HTTPS). It's real, portfolio–ready, and transferable.

Skills you'll gain

- Create and configure S3 buckets for website hosting.
- Understand and apply bucket policies and public access settings.
- Configure CloudFront as a CDN and enable HTTPS.
- Use Git + GitHub to store code and show version history.
- Deploy a simple, responsive landing page.

What you'll build

A responsive single–page portfolio (Hero, About, Projects, Contact) and host it on AWS S3. The site must:

- Use Google Fonts and one icon set (FontAwesome).
- Have a hero with CTA button.
- Contain 3 project cards and a footer.

Deliverables (upload as ZIP or PDF + GitHub link)

1. index.html, styles.css, all assets (images) uploaded to a GitHub repo.
2. Hosted URL (S3 or CloudFront) — show screenshot of CloudFront distribution & S3 bucket policy.
3. Short README.md describing steps you took (commands, links).
4. Optional: a short 60–90s screen recording showing the live site.

Step-by-step (high level)

1. Create project locally (`index.html` , `styles.css` , `assets/`).
2. Initialize Git repository, commit files, push to GitHub (optional).
3. Create an S3 bucket named like `yourname-portfolio` and enable **Static website hosting**.
4. Adjust **Block Public Access** and add a bucket policy to allow `s3:GetObject` .
5. Upload site files to S3 (console or `aws s3 sync` CLI).
6. Test the S3 website endpoint.
7. Create a CloudFront distribution with the S3 bucket as origin; enable **Redirect HTTP to HTTPS** and choose a minimum TLS version.
8. (Optional) Configure Route53/your DNS provider to point a domain to CloudFront.
9. Verify HTTPS works and record final hosted URL.

AWS official guide for S3 static website hosting (step-by-step). [AWS Documentation](#)

Sample Starter Code:

php-template



```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width,initial-scale=1" />
  <title>Your Name – Portfolio</title>

  <!-- Google Fonts -->
  <link
    href="https://fonts.googleapis.com/css2?family=Inter:wght@400;700&display=swap"
    rel="stylesheet"
  >

  <!-- External CSS File -->
  <link rel="stylesheet" href="styles.css">
</head>

<body>
```

```
<!-- Navigation Header -->
<header class="nav">
  <div class="brand">YourName</div>

  <nav>
    <a href="#home">Home</a>
    <a href="#projects">Projects</a>
    <a href="#contact">Contact</a>
  </nav>
</header>

<!-- Hero Section -->
<section id="home" class="hero">
  <h1>Hi – I'm Your Name</h1>

  <p>
    Junior Cloud & DevOps enthusiast.
    I build cloud native apps & deploy them reliably.
  </p>

  <a class="btn" href="#projects">View Projects</a>
</section>

<!-- Projects Section -->
<section id="projects" class="cards">

  <article class="card">
    <h3>Project A</h3>
    <p>Short description. Tech: HTML, S3, CloudFront.</p>
  </article>

  <article class="card">
    <h3>Project B</h3>
    <p>Short description. Tech: Docker, CI.</p>
  </article>

  <article class="card">
    <h3>Project C</h3>
    <p>Short description. Tech: Terraform (optional).</p>
  </article>

</section>
```

```
<!-- Footer -->
<footer id="contact">
  <p>
    © 2025 Your Name —
    <a href="mailto:you@example.com">you@example.com</a>
  </p>
</footer>

</body>
</html>
```

styles.css

```
:root {
  --bg: linear-gradient(120deg,#4facfe,#00f2fe); --card:#fff;
  --accent:#0b69ff; --muted:#7a7a7a;
  font-family:'Inter',system-ui,sans-serif;
}
*{box-sizing:border-box}
body{margin:0;background:var(--bg);color:#111;min-height:100vh}

.nav{
  display:flex;justify-content:space-between;align-items:center;
  padding:1rem 2rem;background:rgba(255,255,255,0.06);
}
.brand{font-weight:700;color:#fff}
.nav a{color:#fff;margin-left:1rem;text-decoration:none}

.hero{
  min-height:60vh;display:flex;flex-direction:column;
  justify-content:center;align-items:center;text-align:center;
  color:#fff;padding:4rem 1rem;
}

.hero h1{font-size:2.4rem;margin:0}
.hero p{color:rgba(255,255,255,0.9);max-width:700px}

.btn{
  display:inline-block;margin-top:1rem;padding:0.6rem 1rem;
  background:#fff;color:var(--accent);border-radius:22px;
  text-decoration:none;font-weight:600;
}
```

```

.cards{
  display:grid;grid-template-columns:repeat(auto-fit,minmax(220px,1fr));
  gap:1rem;padding:2rem;
}
.card{
  background:var(--card);border-radius:12px;padding:1.2rem;
  box-shadow:0 6px 18px rgba(11,105,255,0.08);
}

footer{
  padding:1rem;text-align:center;color:var(--muted);
}

@media(max-width:600px){
  .hero h1{font-size:1.6rem}
}

```

Helpful CLI snippets

bash



sync local site to S3 (replace bucket)

aws s3 **sync** . s3://yourname-portfolio --acl public-read

set bucket policy (example) - use console to craft correct JSON

aws s3api put-bucket-policy --bucket yourname-portfolio --policy file://policy.json

Task-1 Support Resources:

► YouTube Video Tutorials (Highly Recommended)

AWS S3 Website Hosting (Beginner Friendly)

1. Host a Static Website on Amazon S3 (FULL GUIDE)
<https://www.youtube.com/watch?v=K4JBbKx4v40>
2. AWS S3 Website Hosting Step-by-Step (Very Clear Explanation)
<https://www.youtube.com/watch?v=lbzGLX14t4Y>
3. How to Add CloudFront CDN to S3 Website + HTTPS
<https://www.youtube.com/watch?v=Qmz6cf5dtJE>
4. CloudFront Explained for Beginners
<https://www.youtube.com/watch?v=RrKRN9zRBWs>
5. AWS for Beginners — Full Course (FreeCodeCamp)
<https://www.youtube.com/watch?v=ulprqHHWIng>
(Great for interns new to AWS)

■ Free Study Material & Docs

- AWS S3 Static Website Hosting Docs:
<https://docs.aws.amazon.com/AmazonS3/latest/userguide/WebsiteHosting.html>
- CloudFront Developer Guide:
<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/Introduction.html>
- AWS Free Tier (for practice):
<https://aws.amazon.com/free/>